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RESEARCH ARTICLE

HIDS-IoMT: A Deep Learning-Based Intelligent Intrusion Detection System for the Internet of Medical Things

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ABSTRACT The expansion of the Internet of Medical Things (IoMT) has enhanced the accuracy, real-time functionality, connectivity, and intelligence of medical examination practices. However, increased interconnectivity of medical equipment has rendered IoMT networks susceptible to several cyberattacks, particularly the Distributed Denial of Service (DDoS) attacks. Current intrusion detection systems (IDS) are insufficiently focused to address and counteract these advanced threats. Furthermore, due to the dynamic nature of IoMT traffic, IDS has considerable difficulty preserving its current threat detection capabilities. This study presents a hybrid deep learning-based intrusion detection system for IoMT networks (HIDS-IoMT). The proposed model hybridizes the Convolutional Neural Network (CNN) for feature extraction and the Long Short Term Memory neural network (LSTM) for sequence data prediction. We implement the designed IDS on a Raspberry Pi device using a fog computing architecture, enabling decentralized processing closer to IoMT devices, thereby enhancing responsiveness and reducing latency. We evaluate the suggested approach for intrusion detection using the IoTID20 and the Edge-IIoTset datasets. These datasets comprise a substantial and varied assortment of traffic flows from actual DDoS attacks, including SYN floods, UDP floods, HTTP floods, and others. We evaluated our novel methodology against distinct DoS attacks such as DDoS, SQL injection, Vulnerability scanner, Scan Host Port, Mirai, etc. Our proposed model attains an accuracy of 99.92%, a precision of 99.91%, a recall rate of 99.99%, and an F1-score of 99.95%, outperforming the latest in advanced methodologies.

INDEX TERMS Internet of Medical Things, DDoS attacks, intrusion detection system, deep learning, convolutional neural network, long short term memory neural network.

I. INTRODUCTION

The incorporation of medical devices into the Internet of Things (IoT), also referred to as Healthcare 5.0, has led to the emergence of the Internet of Medical Things (IoMT) within the healthcare sector. This innovation has resulted in a new era of patient care and medical practice. The Internet of Medical Things signifies the combination of advanced technologies essential for enhancing health care processes, optimizing patient outcomes, and revolutionizing healthcare services. This extensive and intricate ecosystem comprises several advanced medical equipment and sensors, each carefully engineered to observe, gather, and relay critical healthcare information instantly [1], [2], [3], [4]. The IoMT covers a variety of devices, incorporating wearable health technologies like smartwatches and fitness trackers that assess vital signs and physical activity, as well as implanted medical devices such as pacemakers and insulin pumps that provide critical therapies. Furthermore, the IoMT functions through smooth data interchange, with the primary goal of developing efficient and trustworthy connectivity across a wide range of devices [5], [6], [7].

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However, the rapid expansion and influence of the IoMT in healthcare have rendered it a potential target for malicious actors. The IoMT devices are engineered with limited computing resources, strict power consumption requirements, and frequently restricted memory capacity. As a result, these devices are unable to implement intricate security algorithms, leaving them vulnerable to various security risks like data integrity attacks, denial of service (DoS) attacks, interception, and information leakage [8], [9], [10], [11]. These potential effects may include faulty diagnosis, delayed treatment, or perhaps severe outcomes, including mortality. In this situation, the IDS is crucial for improving the security of the IoMT against malicious attacks. An IDS is an advanced safety system that continuously examines network traffic, carefully analyzing patterns and anomalous behaviors to detect any suspicious or malicious activities. Also, an IDS system should be able to mitigate the effects of intrusions before they can compromise the safety and functionality of IoMT devices [12], [13], [14].

Hence the aim of this research is to build an innovative deep learning-based intrusion detection system for medical IoT networks (HIDS-IoMT). We achieved this by utilizing a hybrid methodology that integrates Convolutional Neural Networks (CNN) with Long Short Term Memory (LSTM) neural networks. We deploy the envisaged IDS on a Raspberry Pi device using a fog computing architecture, enabling decentralized processing closer to IoMT devices, thereby enhancing responsiveness and reducing latency. We trained and assessed using the IoTID20 [15] and the Edge-IIoTset [16] datasets because of their excellent dependability and resemblance to real-world attacks. HIDS-IoMT primarily targets IoMT devices; nevertheless, its core principles and architecture are applicable to diverse IoT applications, including smart cities, Industrial IoT (IIoT), and linked home environments. The following outline summarizes the principal contributions of this paper:

- Analysis of DDoS attacks in fog computing for IoMT networks: This study analyzes the impact of DDoS attacks on IoMT networks inside a fog computing structure, highlighting vulnerabilities and challenges.
- Development of HIDS-IoMT for IoMT networks: We provide a novel hybrid deep learning-based intrusion detection system (HIDS-IoMT) that employs CNN and LSTM models to effectively identify and mitigate threats on IoMT networks.
- Implementation of a modular algorithm for real-time intrusion detection: We implement a modular algorithm designed for real-time intrusion detection in the Internet of Medical Things (IoMT) domain by deploying Raspberry Pi devices in a fog network architecture. This approach reduces latency, ensures cost-effectiveness, and scales easily, making it highly suitable for resource-constrained, latency-sensitive healthcare applications.
- Performance assessment of the proposed intrusion detection: Using feature selection methodologies, the proposed model achieves an accuracy of 99.92%, a precision of 99.91%, a recall rate of 99.99%, and an F1-score of 99.95%.

The present work is structured to systematically examine the development, deployment, and evaluation of the studied HIDS-IoMT scheme. Section II provides a comprehensive analysis of related works, situating our approach within the other structures of intrusion detection research. Section III presents the methodological framework, specifying the architectural structure, the DDoS attack model, and the design of the CNN-LSTM architecture. Section IV details the experimental setup, hyperparameter adjustment, and performance evaluation for both binary and multiclass classifications. Sections V and VI present a detailed analysis, conclusions, and recommendations for future research.

II. RELATED WORKS

The study [17] introduces a new deep learning approach referred to as Pearson-Correlation Coefficient—Convolutional Neural Networks (PCC-CNN). The work is explicitly targeted for internet of things applications, integrating linear-based extractions with Convolutional Neural Networks to execute binary and multiclass classifications. The results indicate that the suggested system surpasses conventional methods, with a detection accuracy of 99.89%.

The research [18] presents an attention-based convolutional neural network (ABCNN) for intrusion detection within the IoT. The ABCNN incorporates an attention mechanism to effectively learn from classes with limited instances. It focuses on essential characteristics and accurately detects undesirable behavior. Through experimental results, the proposed model exhibited outstanding performance across all datasets, with an F1-score of 98.08% and an accuracy of 99.81%, outperforming existing modern approaches.

Alshehri et al. [19] present a self-attention mechanism using the deep convolutional neural network (SA-DCNN). The model aims to monitor industrial internet of things networks and identify malicious behavior. The self-attention mechanism allocates importance to each input characteristic, but the DCNN evaluates these features to assess network behavior. A dual-phase cleaning procedure mitigates data redundancy by addressing both intraclass and cross-class samples. To prevent underfitting, a mutual information-based feature filtering technique evaluates and eliminates characteristics that exhibit a detrimental effect. The IoTID20 and Edge-IIoTset datasets are used to train this model, and simulation results demonstrate that the deployed IDS outperforms several other baseline learning models.

In research [20], the authors proposed an advanced Artificial Intelligence (AI) technique to find intrusions, focusing on finding strange patterns in the way IoT devices behave and adapting quickly to new threats. They incorporate an advanced intrusion identification system for real-time anomaly identification. Experiments and observations reveal that the suggested framework achieves impressive accuracy,

recall, and precision of 98.25%, 96.8%, and 97.45%, respectively, thereby providing a robust solution for the security of IoT ecosystems.

The authors of the study [21] introduce a multi-head attention-based gated recurrent unit (MAGRU) model intended for analyzing industrial internet of things network data and identifying malicious activity. The gated recurrent unit (GRU) focuses on finding IIoT network activity, while the multi-head attention (MA) part improves the model's ability to learn, especially when dealing with small sample classes. MAGRU yielded excellent outcomes, showing that intrusion detection works best in IIoT networks.

In their study, the authors [22] present deep learning-based techniques for detecting network intrusions. They employed real-time traffic datasets, the CICIDS2018, and the Edge-IIoTset for training. The authors evaluated the model's performance by putting it through multi-class classification. It did 100% correctly on the CICIDS2018 dataset and 99.64% correctly on the Edge-IIoTset dataset, showing that it is good at finding threats.

An additional innovative approach is described in [23]. In this research work, the authors propose a novel anomaly-based IDS that integrates machine learning and statistical techniques to detect anomalous patterns that are indicative of DDoS attacks. By exploiting SDN's fundamental programmability, the proposed IDS demonstrates significant adaptability to changing network conditions. The authors emphasize the essential need for rapid detection and an effective response to DDoS attacks, highlighting their capacity to compromise network availability. Simulated scenarios further demonstrate the framework's efficacy in detecting and preventing various DDoS attacks.

The authors in [24] present a review of DDoS attack detection using IDS. The study examines various existing methodologies, highlighting their modification in response to the continuously dynamic nature of DDoS threats. The authors examine diverse detection strategies, such as signature-based, anomaly-based, and hybrid approaches, subsequently evaluating their strengths and limitations. The paper offers insights into various detection methods, enhancing the understanding of IDS mechanisms and assisting researchers in developing effective defenses against a wide variety of DDoS attacks.

Another intrusion detection approach, capable of detecting the presence of cyberthreats in Industrial Control Systems (ICS), is presented in [25]. The paper defines the issues presented by DDoS attacks to vital infrastructures, including industrial control systems. The proposed Network Intrusion Detection System (NIDS) utilizes deep autoencoders to identify complex patterns and distinguish abnormalities indicative of DDoS activity. The autoencoder model is trained on typical network behavior to acquire representative encoding, and deviations from this established pattern activate alerts for potential DDoS occurrences. The authors illustrate the efficacy of the proposed NIDS in identifying DDoS attacks through experimentation and assessment. Furthermore, the use of deep autoencoders demonstrates the ability to accommodate the complexity of ICS network traffic, making it a viable way to improve cybersecurity in critical infrastructures. Thus, it presents a feasible strategy for enhancing cybersecurity in critical infrastructures.

Habeeb and Babu [26] executed a systematic feature selection methodology. They began by examining the link between attributes to discover and eliminate redundancies. Subsequently, they employed a hybrid optimization method that integrates the Whale Optimization Algorithm (WOA) [27] with the Genetic Algorithm (GA). The model attained an accuracy of 99.5%.

Mohy-eddine et al. [28] use the K-nearest neighbors (KNN) on the BoT-IoT dataset to enhance the detection rate. They combine principal component analysis, univariate statistical tests, and genetic algorithms to identify the ten most pertinent characteristics. The experiments and the observations reveal the performance of the proposed technique is proved to have 99.99% as F1-score, with a detection duration of 57.73 seconds.

The study [29] proposed a new IDS for Wireless Sensor Networks (WSNs). The authors apply the architecture of convolutional neural networks to evaluate network traffic patterns and identify intrusion attempts. The CNN model is trained and supervised with labeled datasets to classify network traffic as either normal or malicious. In the context of WSNs, obtaining adequately structured data might be difficult due to the networks' limited resources and inherent constraints. Moreover, CNNs are computationally intensive and necessitate substantial processing power and memory resources for both training and inference.

In [30], the authors introduce a novel smart and automated IDS that aims to determine and categorize malicious activity within the internet of things networks. The suggested IDS can categorize network traffic into five classifications: normal, Mirai attack, denial of service (DoS) attack, scan attack, and man-in-the-middle (MITM) attack. In [30], training of the proposed framework is carried out on the IoTID20 dataset and the authors achieved detection accuracy that varied from 99.40% to 100%.

Additional study [31] presenting hybrid approaches incorporating both the Gorilla Troops Optimizer (GTO) [32] and the Bird Swarm Algorithm (BSA) [33]. The suggested GTO-BSA possesses a significant ability to identify the appropriate locations with the best solutions.

The study [34] introduced a lightweight model to improve the detection of IoT botnet attacks. They used the Jaccard Index [35] and the Pearson Correlation Coefficient (PCC) [36] to determine the most significant and relevant features. They gathered a dataset from three categories of IoT devices for their research and were able to obtain an accuracy rate of 99.70%.

Another detection approach was proposed by Ullah et al. [37]. The proposed approach is based on Convolutional Neural Networks (CNN) to detect real-time

IoT network attacks, and they employed transfer learning methodologies to create models qualified in both binary and multi-class classifications of network traffic.

In [38], researchers introduce a novel approach combining AlexNet and adversarial training to improve the accuracy of pneumonia diagnosis. The AlexNet model, pre-trained on ImageNet, uses transfer learning to extract features from chest X-rays and is subsequently fine-tuned on a labeled dataset of X-ray images categorized as normal and pneumonia cases. Adversarial training produces synthetic X-rays with minor perturbations, improving the model's resistance and generalization abilities. The suggested method attained a validation accuracy of 98.28%, surpassing the conventional AlexNet without adversarial training. Evaluation metrics, such as sensitivity, specificity, and F1 score, underscore the model's capacity to reduce false positives and false negatives.

In [39], the authors explore the application of a novel fusion of two deep learning models, ResNet-50 and Inception-V3, for the classification of brain tumors into either benign or malignant. The combined architecture depends on ResNet-50's deep convolutional capability and Inception-V3's efficacy in mitigating overfitting, facilitating the precise classification of intricate picture patterns. Metrics including sensitivity, specificity, and F1-score highlight the model's robustness, especially with an accuracy of 98.67%.

Ibriahim et al. [40] present a comprehensive methodology for identifying β-thalassemia carriers using a late fusion machine learning model. Leveraging logistic regression, Naïve Bayes, decision tree, and neural network algorithms, the study integrates their outputs through a fuzzy inference system to enhance prediction accuracy. The proposed model achieved a superior accuracy of 96% and outperformed prior approaches in efficiency and reliability. This novel technique combines individual algorithm strengths, minimizing errors and improving precision.

The research study [41] investigates the application of transfer learning with explainable artificial intelligence (XAI) to enhance ocular disease diagnosis. The proposed model utilizes EfficientNet as a base and is fine-tuned using a dataset from the Ocular Disease Intelligent Recognition (ODIR) repository, attaining an interesting accuracy of 95.74%. XAI methodologies, particularly Local Interpretable Model-Agnostic Explanations (LIME), were employed to augment transparency in decision-making, thereby reducing the limitations of conventional "black-box" AI models. This approach allows clinicians to precisely analyze diagnostic outcomes, thereby improving trust in AI-supported medical care.

The study [42] investigates the challenges in identifying diabetic retinopathy and emphasizes the need for accuracy and clarity in interpreting different diagnoses. The study illustrates how traditional AI models typically function as opaque systems, providing limited information regarding their decision-making processes. This deficiency in openness constrains healthcare practitioners' capacity to trust and effectively employ these models. The research presents a novel scheme that combines convolutional neural networks (CNNs) with understandable artificial intelligence (XAI) techniques. This integration enhances transparency by elucidating the reasoning underlying forecasts.

An additional approach is presented in [43], where the authors address the significant difficulty of protecting IoT networks employed in healthcare applications, particularly for individuals with disabilities. The proposed scheme presents a hybrid intrusion detection system (IDS) that utilizes preprocessing approaches, including data normalization and feature selection via the Pearson Correlation Coefficient (PCC). Subsequently, classification is performed with a Support Vector Machine (SVM) model. The proposed work achieves impressive outcomes with an accuracy of 99.3%, a precision of 99.1%, and an F1-score of 99.25% on the KDD99 benchmark dataset.

III. METHODOLOGY

This section outlines the deployed scheme for an intrusion detection system. Indeed, the primary goal of an IDS is to detect and distinguish between distinct attacks in medical IoT networks. To accomplish this goal, the design must incorporate numerous collaborative elements and utilize adaptive and flexible approaches to ensure the highest degree of accuracy. In subsequent sections, we will further discuss the hybrid configuration of our suggested IDS, which incorporates several characteristics, placement strategies, and detection approaches.

A. ARCHITECTURAL STRUCTURE AND ACTUATORS

This part specifies the architecture of the deployed model and its components, aiming to develop a strong and sophisticated network security framework that can successfully counter potential threats. The suggested architecture has three essential device categories: (i) edge devices, (ii) fog nodes, and (iii) a central server, as depicted in Figure 1. Every layer possesses a specific function that enhances the system's overall performance and security.

The underlying layer, dedicated to healthcare wearable devices, comprises a collection of sensors that facilitate the monitoring of vital signs, medication compliance, and other essential health metrics. These devices, mostly comprising IoT sensors and actuators, play a crucial part in monitoring the health, behavior, and location of patients using wearable sensors. Edge devices, strategically located within the healthcare environment, collect real-time data that is essential for patient care and monitoring. Nonetheless, due to their constraints in storage and processing capabilities, these devices rely on efficient communication pathways with higher layers to ensure timely data processing and service delivery.

The fog layer, situated between the cloud and edge layers, quickly processes the data gathered from edge devices. This makes it easier for these devices to respond quickly to critical health events by reducing the amount of work they have to do. The fog layer comprises many fog nodes (FN_1, FN_2, ...,

TABLE 1. Overview of deep learning methodologies employed for anomaly-based IDS in IoT networks.

Ref	Year	Dataset	Detection method	Evaluation metrics	Benefits	Limitations
[17]	2023	NSL-KDD, CICIDS-2017, and IOTID20 datatsets	Anomaly-based detection	Accuracy, Precision, Recall, F1-Score, and False Positive Rate (FPR)	Effective at detecting unknown or zero-day attacks. Can handle large IoT network traffic efficiently	High false-positive rates in certain scenarios. Performance may degrade with highly imbalanced datasets
[18]	2024	Edge-IoTset, IoTID20, ToN_IoT, and CIC-IDS2017 datasets	Anomaly-based detection	Accuracy, Precision, Recall, F1-Score, Area Under the Curve (AUC), and False Positive Rate (FPR)	High detection accuracy for both known and unknown attacks. Reduced computational complexity compared to traditional CNN methods.	Higher computational resource requirements for real-time detection
[19]	2024	IoTID20 and Edge-IIoTset datasets	deep convolutional neural network	Accuracy, Precision, Recall	Self-attention mechanism improves feature extraction and model performance	Requires high computational resources, especially in real-time scenarios
[20]	2024	IoTID20 and N-BaIoT datasets	Anomaly-based detection	Accuracy, Precision, and Recall	Enhanced security by combining trust management with ML for dynamic and adaptive detection	Computational complexity due to the integration of multiple modules
[21]	2023	Edge-IIoTset and MQTTset datasets	Anomaly-based detection	Accuracy, Precision, Recall, and F1-Score	GRU helps reduce computational overhead compared to other RNNs like LSTMs	Model performance might be sensitive to dataset size and quality
[22]	2023	CICIDS2018 and Edge-IIoTset datasets	Deep Neural Network	Accuracy	Capable of capturing both spatial and temporal patterns in data	Requires significant computational resources for training and inference
[23]	2021	CICDoS2017 and CICD-DoS2019 datasets	Anomaly-based detection	Accuracy, F1-score, and classification time, Latency detection	Lightweight and early detection	Specific only for DDoS attacks
[25]	2023	Cyber-security ICS dataset	Deep autoencoder	True Positives (TP), False Positives (FP), True Negatives (TN), False Negatives (FN), Precision, Recall, TPR, FPR, G-mean, F1-Score, AUC-ROC, and AUC-PR	Performed well when trained on Intrusion Detection System (IDS) datasets	Lack to be adopted in real-time systems due to their incapacity to generalize and be flexible to different traffic profiles and networks
[29]	2023	UNSW-NB15 dataset	Convolutional Neural Network (CNN)	Accuracy, detection precision, specificity, and F1-score	Improved detection accuracy (99.4%), Real-time detection	high computational requirements, scalability issues in real-time scenarios

TABLE 1. *(Continued.)* **Overview of deep learning methodologies employed for anomaly-based IDS in IoT networks.**

[30]	2022	IoTID20 dataset	Anomaly-based detection	accuracy, precision, recall, F1-score, and specificity	High detection accuracy (99.4% - 100%)	Limited dataset scope, scalability issues in large-scale IoT environments
[38]	2023	ImageNet dataset	AlexNet deep learning architecture	Accuracy, precision, sensitivity, AUC-ROC, and F1-score	Higher detection accuracy, improved robustness to small perturbations	Potential class imbalance
[39]	2023	253 MRI images	ResNet-50 and Inception-V3 architectures	Accuracy, specificity, and sensitivity	High accuracy in tumor classification (98.67%)	Small dataset size may limit generalizability
[40]	2024	5,066 records from the Punjab Thalassemia Prevention Program (PTPP)	fuzzy inference system	Accuracy, specificity, and sensitivity	High prediction accuracy and reliability (96%)	Relies on a specific dataset, limiting generalizability
[41]	2024	Ocular Disease Intelligent Recognition (ODIR) dataset	Transfer learning using EfficientNet	Accuracy, Miss Rate	High accuracy in predicting ocular diseases	Lack of real-time application capability
[42]	2024	A dataset of retinal images	Convolutional Neural Networks (CNNs) integrated with Explainable AI (XAI) techniques	Accuracy, Miss Rate	Enhances diagnostic accuracy	Relatively small dataset limits generalizability
[43]	2022	KDD99 benchmark dataset	Support Vector Machine (SVM) classification	Accuracy, precision, and F1-score	High accuracy and precision in detecting intrusions	Limited to the KDD99 dataset
Our	2024	IoTID20 and Edge-IIoTset datasets	Convolutional Neural Network (CNN) - Long Short Term Memory (LSTM)	Accuracy, precision, recall rate, and F1-score	High detection accuracy (99.87%), improved precision rate (98.94%)	Long testing periods still pose a concern for deployment on resource-constrained IoMT devices

FN_n) that serve as intermediaries for data processing and transmission. In contrast to edge devices, fog nodes have superior processing capabilities, allowing them to manage complicated tasks like advanced data analysis and anomaly detection. This study emphasizes the practical application of Raspberry Pi as an effective fog node, situated as the first line of defense close to the IoT layer. Raspberry Pi mitigates security threats early by processing data near the source, thereby substantially decreasing the computing and networking loads on the cloud layer. The fog layer maintains consistent connections to the cloud through cloud gateways (CGs) and base stations (BSs), facilitating efficient communication and resource management. This architecture ensures continuous data transfer across multiple network layers and facilitates consistent updates and synchronization between layers, essential for maintaining the network's overall security and efficiency. Our deep learning HIDS-IoMT is built into every fog node.

The uppermost layer, commonly known as the cloud layer, offers resilience and scalability for the storage of significant data values. It serves as a centralized repository that maintains up-to-date information on the behavior of diverse IoT devices. The cloud layer sends frequent updates to the lower layer IoT devices, ensuring the accuracy of their behavioral records, regardless of whether the devices are operating normally or facing an attack.

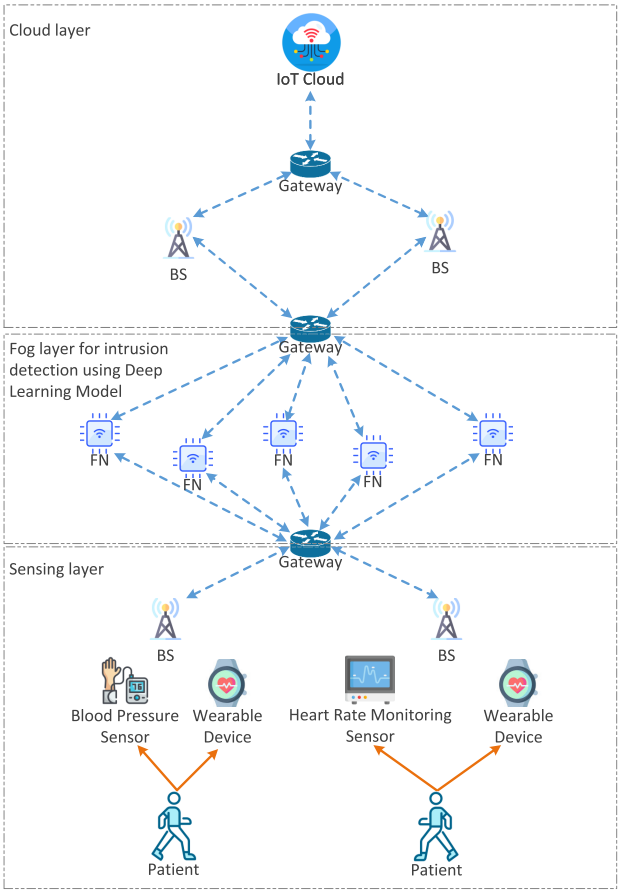


FIGURE 1. Architectural structure and actuators of the proposed model.

B. DDOS ATTACK MODEL

This section analyzes the security consequences of a healthcare system in which a malicious node attempts to violate the communication channel to compromise wearable sensor devices, including heart rate sensors, breathing rate sensors, and activity level sensors. Figure 2 depicts how the attacker infiltrates the healthcare system by injecting malicious messages that degrade or disable healthcare services. Compromised devices employ multiple techniques to harm fog nodes in the intermediary layer, including overflowing the network with a significant amount of requests that disrupt its functionality.

In this regard, an attacker gains control of an apparently legitimate but compromised wearable sensor device, such as a heart rate sensor, within the lowest layer. This malicious node tries to compromise healthcare services by injecting fraudulent messages into the communication channel between healthcare devices and fog nodes. These harmful communications appear to be genuine data, but they contain significant distortions that interfere with the processing of patient health measurements and deceptive requests that exceed the fog layer's processing capabilities. The fog nodes, tasked with processing healthcare data from various devices, experience congestion, leading to delayed or erroneous processing of critical patient information like heart rate, respiratory rate, and activity levels. This interruption may result in significant consequences, including postponed medical treatment or even misdiagnosis.

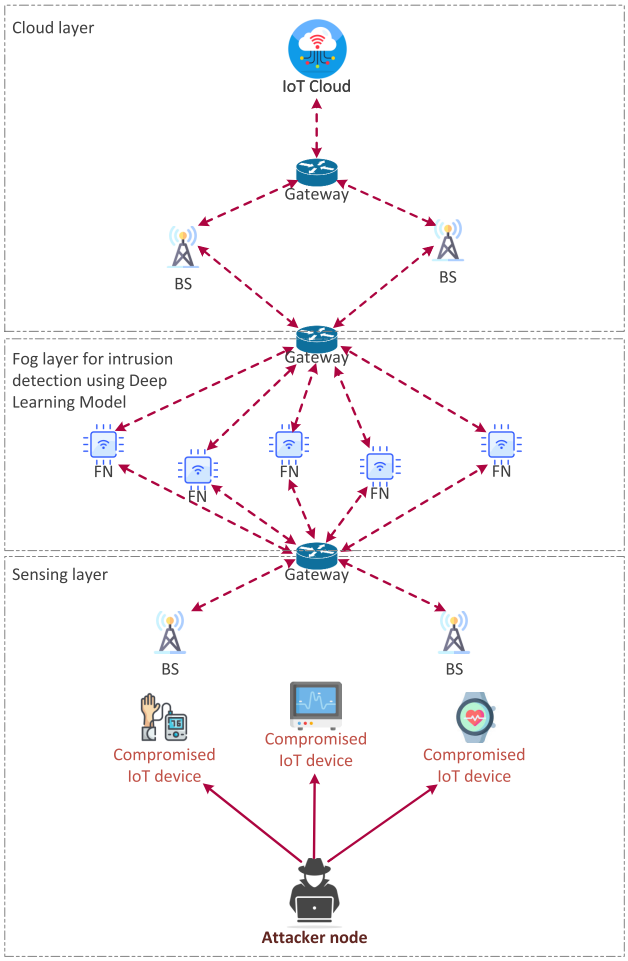


FIGURE 2. DDoS attack model.

C. DESIGN OF THE CNN-LSTM ARCHITECTURE

The integration of deep learning networks, especially CNN and LSTM, into the Internet of Medical Things signifies a crucial paradigm in intrusion detection systems [44], [45], [46], [47]. It simplifies the acquisition of complex attributes and data patterns necessary for identifying network vulnerabilities. CNN and LSTM demonstrate distinctive features that make them especially appropriate for their specific functions within the CNN-LSTM system. CNN performs feature extraction from data. The capacity to process extensive traffic data enables CNN to discern significant aspects pertinent to large-scale network environments. The CNN architecture incorporates three fundamental layers: the convolutional layer, the pooling layer, and the fully connected layer. The convolution layer emphasizes feature extraction from the input data, while the pooling layer samples the extracted features. The fully connected layer integrates the retrieved data and employs a classifier to produce precise classification outcomes.

LSTM, on the other hand, is an improved alternative to the conventional recurrent neural network (RNN) methodology [48]. Its advantage resides in its capacity to collect long-term dependencies and contextual information, rendering it especially appropriate for the dynamic and evolving nature of IoMT networks. In contrast to conventional RNN units, LSTM utilizes a series of gate functions to manage feedback, allowing it to eliminate short-term errors while preserving long-term features. This method enables LSTM to effectively manage long-term sequence data and substantially enhance the precision of intrusion detection. As illustrated in Figure 3, the data processing sequence in LSTM consists of 3 gates: the input gate (i_t), the output gate (o_t), and the forget gate (f_g) [49], [50], [51], [52].

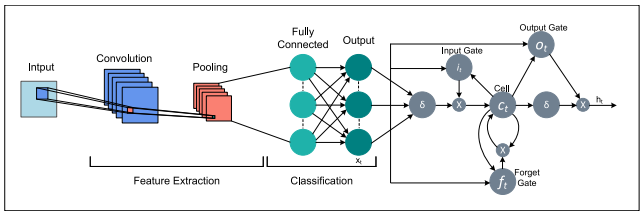


FIGURE 3. System architecture of the proposed CNN-LSTM model.

D. DESCRIPTION OF HIDS-IOMT PROCEDURE

Algorithm 1 presents the pseudocode for the HIDS-IoMT approach. This procedure is initiated upon the receipt of a message from an adjacent node (fog or sensor node). Initially, the proposed algorithm examines the message and subsequently performs the required routing management activities. If a message is received from a neighbor suspected of being an attacker, the packet is rejected, and the alarm mechanism is initiated against the suspicious node.

This research introduces a threshold, denoted as τ, which signifies the time gap between the reception of successive messages from the same nearby node. When this interval is equal to or less than τ, there is a suspicion that the adjacent node could have suspicious intentions. If the interval exceeds τ, the neighbor is considered non-malicious. The value of τ, as stated by Thulasiraman et al. [53], is established at 500 milliseconds.

When a node receives a message from a non-existent node in the neighborhood table, it adds the IP address of the message's source node into its neighborhood table. Subsequently, the node initiates a call to the intrusion detection system procedure (HIDS-IoMT) to figure out whether the message constitutes an attack or not. Every 30 seconds, the ready variable triggers the IDS method. The HIDS-IoMT method executes the following functions:

1) Initialization and configuration: This method involves the creation and initial setup of the neighboring and malicious tables.
2) Prompt detection of malicious nodes: The HIDS-IoMT procedure is engineered to quickly detect the origin of messages, specifically targeting messages from prohibited list nodes. In this scenario, the system promptly rejects the incoming message and activates a corresponding alarm.

Algorithm 1 Pseudo-Code to HIDS-IoMT

procedure HIDS-IoMT
  $\lambda_{now} \leftarrow localtime()$ ▷ Get the Local Time System
  if ($Entry_{Status}$ = FALSE) **then** ▷ If prohibited list and neighbor tables are empty?
    Call *Setup_adjacent_table* Procedure
    Call *Setup_malicious_table* Procedure
    $Entry_{Status} \leftarrow TRUE$
  end if
  for i ← 1 to N_{Denied} **do**
    if (($\Gamma[\Phi.SrcIP_{denied} = SrcIP_{recv_msg}]$ AND ($\Gamma[v.status]$ = TRUE)) **then**
      Received message rejected
      Call *Generate_corresponding_alert* Procedure
      return
    end if
  end for
  $\alpha \leftarrow FALSE$ ▷ Set the node's neighbor table entry to "FALSE"
  for j ← 1 to $Count_{neigh}$ **do**
    if ($\Phi[\Upsilon.Src_{Sender} = SrcIP_{recv_msg}]$) **then**
      $\alpha \leftarrow TRUE$
      $\Phi[\Upsilon.T_{Previous}] \leftarrow \Phi[\Upsilon.T_{Actual}]$
      $\Phi[\Upsilon.T_{Actual}] \leftarrow \lambda_{now}$
      $\Phi[\Upsilon.Count_{Msg}] \leftarrow \Phi[\Upsilon.Count_{Msg}] + 1$
    end if
  end for
  if ($\alpha = FALSE$]) **then** ▷ Verify that the node is not present in the neighboring table
    for k ← 1 *To* $WearNode_{max}$ **do**
      if ($\Phi[\Upsilon.Src_{Sender}] = NULL$]) **then** $\Phi[\Upsilon.T_{Previous}] \leftarrow \Phi[\Upsilon.T_{Actual}]$
$\Phi[\Upsilon.T_{Actual}] \leftarrow \lambda_{now}$

        $\Phi[\Upsilon.Src_{Sender}] \leftarrow SrcIP_{DIO}$
        $\Phi[\Upsilon.Count_{Msg}] \leftarrow \Phi[\Upsilon.Count_{Msg}] + 1$
        $Count_{neigh} \leftarrow Count_{neigh} + 1$

      end if
    end for
  end if
  if ($ready = TRUE$]) **then**
    Call *Check_malicicous_node* Procedure

    $ready \leftarrow FALSE$

  end if
end procedure

Algorithm 2 Pseudo-Code to Setup Adjacent Table

```
procedure Setup_adjacent_table(Φ, ϒ)
    for i ← 1 to WearNode_max do
        Φ[ϒ.Src_Sender] ← NULL
        Φ[ϒ.T_Previous] ← 0
        Φ[ϒ.T_Actual] ← 0
        Φ[ϒ.Count_Msg] ← 0
    end for
end procedure
```

Algorithm 3 Pseudo-Code to Setup Malicious Table

```
procedure Setup_malicious_table(γ, ν)
    for i ← 1 to WearNode_max do
        γ[ν.SrcIP_denied] ← NULL
        γ[ν.Count_detection] ← 0
        γ[ν.Status] ← FALSE
    end for
end procedure
```

Algorithm 4 Pseudo-Code to Remove an Entry From a Table

```
procedure Remove_neighbor_table_entry(index, Φ, ϒ)
    Φ[ϒ_index.Src_Sender] ← NULL
    Φ[ϒ_index.T_Previous] ← 0
    Φ[ϒ_index.T_Actual] ← 0
    Φ[ϒ_index.Count_Msg] ← 0
end procedure
```

3) Adjacent table maintenance: The maintenance of the neighbor table is an essential part of the HIDS-IoMT procedure's responsibilities. This process involves adding new entries to the neighbor table after discovering new adjacent nodes.
4) Periodic inspection of malicious neighbors: We call the HIDS-IoMT procedure every 30 seconds to detect possibly suspicious nodes in the adjacent table. The process employs a series of procedures, including analyzing anomalies in the neighbor database and examining abnormal patterns in received messages, to identify malicious neighbors.

E. DESCRIPTION OF AUXILIARY PROCEDURES

Setup_adjacent_table, *Setup_malicious_table*, and *Remove_adjacent_table_entry* are supplementary procedures utilized within the HIDS-IoMT approach.

The *Setup_adjacent_table* procedure initializes the neighbor table with initial values upon node activation. Algorithm 2 outlines the pseudocode for the *Setup_adjacent_table* procedure. This procedure guarantees the proper initialization of neighbor table entries with default values. Similarly, the pseudocode for the *Setup_malicious_table* procedure, as presented in Algorithm 3, initializes the malicious table. The primary goal of the *Remove_adjacent_table* entry procedure is to eliminate a particular entry from the neighbor table, as demonstrated in Algorithm 4.

The fundamental HIDS-IoMT algorithm exhibits a time complexity of $O(WearNode_{max})$, as it traverses the prohibited list and nearby tables while updating node entries. The Setup Adjacent Table and Setup Malicious Table algorithms, responsible for initialization, each exhibit $O(WearNode_{max})$ time complexity and require $O(WearNode_{max})$ space to store node information. Similarly, the Remove Neighbor Table Entry algorithm operates in $O(1)$ time, as it resets a single table entry. These complexities confirm the algorithms' scalability for large-scale IoT networks. Table 2 presents various symbols together with their corresponding definitions employed in the proposed intrusion detection system.

F. DATASET SELECTION

Datasets are an essential component in the field of machine learning. It is essential to get or generate a dataset that accurately represents real-world scenarios or occurrences for effective training and validation [54], [55], [56]. Our intrusion detection system is developed and evaluated using the IoTID20 and the Edge-IIoTset datasets. The IoTID20 dataset has been gathered from home IoT networks to aid in the identification of cyber-attacks. The primary benefit arises from the integration of current communication data with novel samples; hence, it improves the capacity to detect network breaches. The IoTID20 dataset consists of 625,783 samples, with 40,073 categorized as normal and 585,710 grouped into four distinct categories of attacks. Additionally, these four categories of attacks are further separated into eight subcategories. The simulated attacks are classified into four distinct groups: Denial of Service attack (DoS), Mirai Botnet, Scanning, and Man-in-the-Middle on Address Resolution Protocol:

1) DoS Attack: In a denial-of-service attack, a malicious source targets a single endpoint or node, overwhelming the server with requests to disrupt services and prevent other devices from accessing it.
2) Mirai Botnet: Here, the attacker seeks to compromise the victim's software, aiming to convert it into a bot within a network of remotely controlled devices.
3) Scanning: The attacker gathers data by probing devices and potentially modifies information during the process by leveraging hardware to scan for vulnerabilities.
4) MITM (Man-in-the-Middle) via ARP Spoofing: This attack involves the attacker intercepting communication between two nodes and manipulating the traffic as it passes between them, a common approach in cyberattacks.

The Edge-IIoTset dataset [16] is a recently created dataset that emphasizes IoT cybersecurity, encompassing IoT and IIoT network communication instances collected from a real-world testbed comprising various devices, sensors, and cloud/edge settings. The Edge-IIoTset dataset outlines 62 features and investigates 14 attack scenarios. The 14 attacks are

TABLE 2. Parameters and definitions.

Parameters	Definitions
$WearNode_{max}$	Maximum number of wearable nodes in the network.
$\Gamma \leftarrow [1, \ldots, WearNode_{max}]$	Wearable Nodes Prohibited list Table
$\Phi \leftarrow [1, \ldots, WearNode_{max}]$	Wearable Nodes Neighbor Table
$SrcIP_{denied}$	The prohibited listed node's IP address.
$Count_{detection}$	The cumulative instances in which the wearable device has been recognized as a possible threat
$Status$	The status of the prohibited listed wearable node, where FALSE indicates it's suspected and TRUE indicates it's permanently blocked
Src_{Sender}	Indicates the source IP address of the message
$T_{Previous}$	Denotes the time of the last received message
T_{Actual}	Reflects the timestamp of the most recent message reception
$Count_{Msg}$	Indicates the total number of messages received from that neighbor up to the current time
$\Upsilon_i \leftarrow [< Src_{Sender}, T_{Previous}, T_{Actual}, Count_{Msg} >], i = 1 \ldots WearNode_{max}$	The structure of a neighbor table node entry
N_{Denied}	The number of the prohibited listed wearable nodes
$Count_{neigh}$	The total number of entries in the neighbor table
$Entry_{Status}$	Flag to verify the initialization status of the neighbor table and prohibited list table
λ_{now}	The current time according to the system clock
α	Flag to verify whether the wearable node is present in the neighbor table
$ready$	Global variable indicating the readiness of the IDS to initiate the procedure for detecting attackers in the neighboring table. It is set to TRUE by the legitimate node every 30 seconds
$SrcIP_{recv_msg}$	The IP address source of the message

TABLE 3. Label details of IoTID20 and Edge-IIoTset datasets.

Binary label distribution	IoTID20 Instances	Edge-IIoTset Instances
Normal	40, 073	1,615,643
Anomaly	585,710	603,558

TABLE 4. Encoding.

Category	Numerical Value
Normal	0
DoS	1

categorized into 5 distinct threat types: Distributed Denial of Service (DDoS), Information Gathering, Man-in-the-Middle, Injection, and Malware attacks. The dataset comprises 2,219,201 instances, with 1,615,643 being grouped as legal and 603,558 representing 14 different attacks. Table 3 highlights the amount of normal and malicious instances within the IoTID20 and Edge-IIoTset datasets.

G. DATA PREPROCESSING

Preprocessing is crucial to optimize dataset compatibility with machine learning and deep learning models, as it improves result quality. This study applies several preprocessing steps, including data preparation, feature filtering, data normalization, and dividing the dataset into training and testing sets.

1) DATA CLEANING MODULE

The IoTID20 dataset has 625,783 samples, whereas the Edge-IIoTset dataset contains 2,219,201 entries related to different types of attacks. This substantial volume of data may encompass missing, misleading, and incomplete information. Therefore, we implemented data cleaning on the IoTID20 and the Edge-IIoTset datasets during the data pre-processing phase to meet data quality standards. We eliminated the irrelevant indices and excluded the missing and *NaN* values.

2) FEATURES ENCODING MODULE

The IoTID20 and the Edge-IIoTset datasets have many categorical variables that needed to be turned into numbers so they could be used with deep learning models that are meant to predict how network activity will behave. As shown in Table 4, we employed the label encoding technique to achieve this. This approach allocates a distinct numerical value to each group of values within an attribute. We chose this method because of its superior efficiency in memory utilization and processing power. Also, the label encoding method does not change the dimensionality of the datasets and does not increase the time costs of the classifier training [57], [58].

3) FEATURE SELECTION MODULE

Feature selection is the procedure for removing irrelevant and unproductive features to enhance the learning capacity and predictability of classifiers. The IoTID20 and the Edge-IIoTset datasets include variables that limit model generalization and have no significant effect on the proper classification of the target class. The Particle Swarm Optimization (PSO) algorithm is frequently employed to identify the most important and distinctive features from the input dataset [59]. Thus, PSO not only minimizes computational costs but also reduces training duration and memory usage, positively affecting the system's energy consumption. The primary steps of the PSO algorithm are summarized in Algorithm 5. It begins by defining N as the population size of particles and randomly initializing their positions and velocities. The position and velocity define the particle's state within the solution space, as illustrated by the following equations:

$$p_i = \{p_{i1} + p_{i2} + p_{i3}, \ldots p_{iS}\}, \quad (1)$$

$$vp_i = \{v_{i1} + v_{i2} + v_{i3}, \ldots v_{iS}\}, \quad (2)$$

where S denotes the entire search space for the particles. Algorithm 5 demonstrates that the initial value, P_{best}, signifies the optimal local fitness value reached by each particle, whereas the subsequent value, H_{best}, denotes the maximum fitness value achieved by any particle globally. The updated velocity and position values are calculated using Equation 3 and Equation 4, respectively.

$$v_{id}^{j+1} = w * v_{id}^{j} + c_1 * rand_{1i}(P_{id} - p_{id}^{j}) + c_2 * rand_{2i} * (H_{gd} - p_{id}^{j}) \quad (3)$$

$$p_{id}^{j+1} = p_{id}^{j} + v_{id}^{j+1} \quad (4)$$

where d represents the dimension, and j denotes the iteration variable within the search space. The values c_1 and c_2 represent the acceleration, while the variable w signifies the velocity. The P_{id} and H_{gd} represent the P_{best} and H_{best} values within the search space, respectively, whereas $rand_{1i}$ and $rand_{2i}$ represent random values that are uniformly distributed within the range of 0 and 1. The iterative process in Algorithm 5 proceeds until it attains optimal criteria or designated values. The Edge-IIoTset dataset comprises 61 features, while the IoTID20 dataset encompasses 83 features. We selected 46 features from the Edge-IIoTset and 62 features from the IoTID20 dataset.

Algorithm 5 Feature Selection Using Principal Swarm Optimization (PSO) Technique

Input: Initialize parameter $X_{counter}^{t}$ as fitness function, Total population P_{i_best}, and N Total number of particles
Output: Selected features

```
while Optimal features not found do
    for counter ← 1 to N do
        if fitness X^t_counter > fitness P_i_best then
            Update P_i_best ← X^t_counter
            if fitness X^t_counter > H_best then
                H_best ← fitness X^t_counter
            end if
        end if
        Update velocity of the particle (Equation 3)
        Update position (Equation 4)
        Select next particle in the population
    end for
    Repeat step 1 until the convergence
end while
```

4) DATA OVERSAMPLING MODULE

Training a model using an imbalanced dataset presents greater challenges than training using a balanced dataset. This, in turn, diminishes the effectiveness of the classification model and leads to a rise in false positives. This results in difficulty in preventing biased predictions favoring the majority class because of the limited availability of minority class samples. Figure 4 and Figure 6a illustrate the data imbalance problem of the IoTID20 dataset using histogram analysis, displaying the sample counts for different attack types and their subcategories. After the cleaning process, the IoTID20 dataset includes 585,342 attack samples and 40,073 samples of normal traffic data. Figure 4 counts for the attack subtypes are as follows: Mirari Ack Flooding (55,124), Mirai Host Brute Force (121,178), Mirai HTTP Flooding (55,818), and Mirai UDP Flooding (183,189). Likewise, the Scan Host Port subtype comprises 22,192 samples, the Scan OS Port contains 53,073 samples, DoS Syn Flooding includes 59,391 samples, and MITM ARP Spoofing has 35,377 samples.

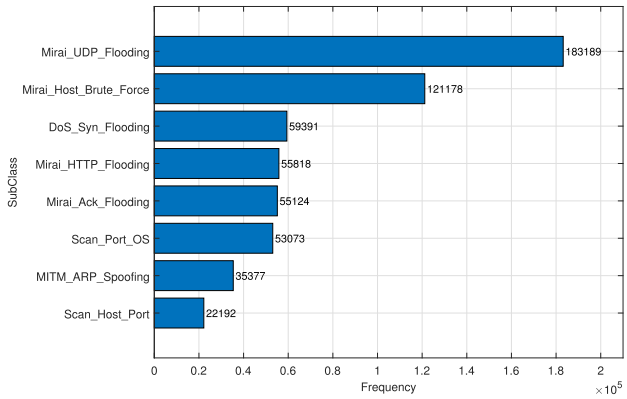


FIGURE 4. Class imbalance issue within IoTID20 dataset.

As with the IoTID20 dataset, after the cleaning process, the Edge-IIoTset comprises a total of 1,909,671 instances. Among these, 1,363,998 occurrences are categorized as normal, whereas the remaining 545,673 instances correspond to 14 distinct attacks. Figure 5 and Figure 6b illustrate a comprehensive distribution of the Edge-IIoTset dataset.

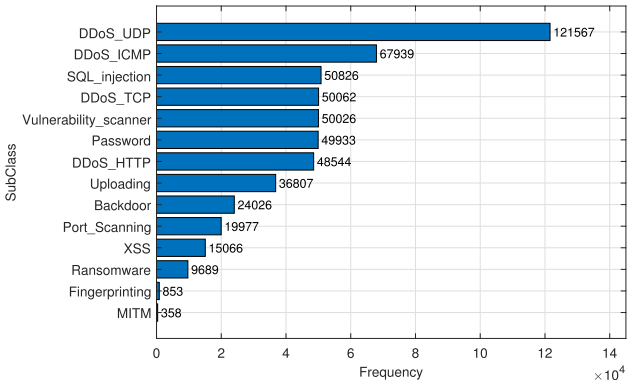
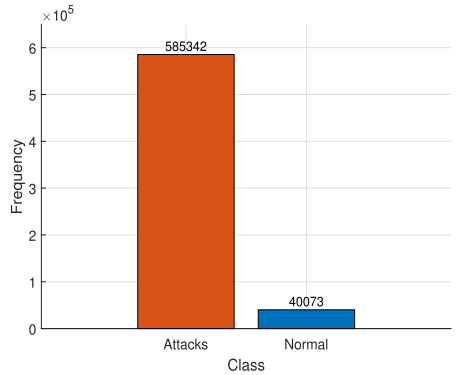
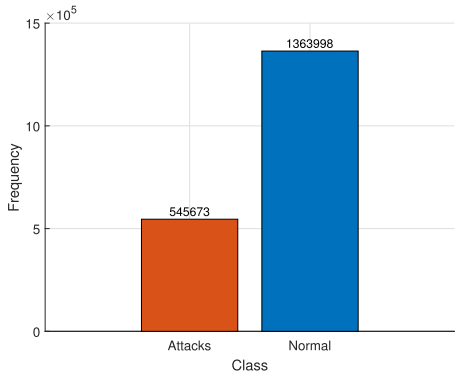


FIGURE 5. Class imbalance issue within Edge-IIoTset dataset.

In view of the extreme imbalance of classes, numerous techniques have been proposed to tackle this issue within the datasets, especially to oversample cases of the minority classes. The Synthetic Minority Over-sampling Technique (SMOTE) mitigates class imbalance by creating synthetic samples of minority classes using interpolation [60], [61], [62]. SMOTE mitigates model bias toward majority classes by augmenting representation, resulting in enhanced accuracy, particularly in fields such as medical diagnostics and



(a) IoTID20 dataset before data augmentation



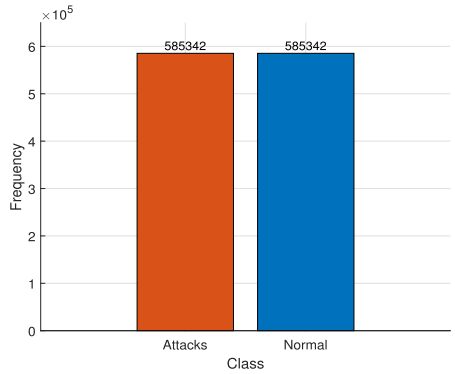
(b) Edge-IIoTset dataset before data augmentation

FIGURE 6. Class imbalance within IoTID20 and Edg-IIoTset datasets: binary classification.

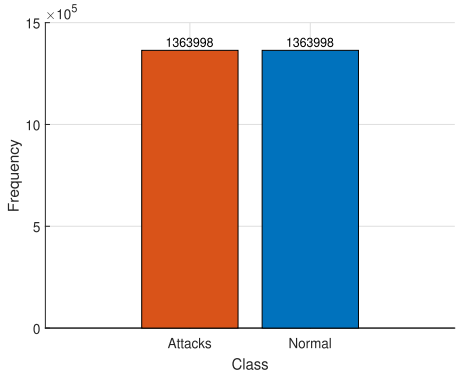
intrusion detection. This paper conducts two distinct experiments aimed at addressing the problem of imbalanced datasets.

The first instance of data balancing refers to the binary classification between regular data samples and instances of intrusion attacks. The sample analysis indicates that normal samples in the IoTID20 dataset constitute approximately 7% of the total, whereas attack samples account for 93% of the dataset's population. The SMOTE approach balances the binary samples between normal and attack, resulting in 1,170,684 instances within the IoTID20 dataset and 2,727,996 instances within the Edge-IIoTset dataset. Figure 7 illustrates the histogram of the resulting distribution. We partitioned the datasets into training and testing sets in an 80% to 20% ratio, respectively. We further partition the testing set into a final testing set and a validation set, each accounting for 50% of the total.

In the second scenario of our study, issues related to class imbalance appeared in the multiclass categorization of intrusion subtypes. For example, as illustrated in Figure 4, the Host Port attack comprises approximately 22,192 instances, which represents about 3.5% of the dataset, whereas the UDP Flooding attack encompasses 183,189 samples, representing 29.3% of the whole dataset. Comparable differences are present among other attack subtypes, particularly when their sample sizes are significantly smaller than those of UDP Flooding. This illustrates a scenario where an imbalanced dataset could negatively impact the performance of the intrusion detection system. Therefore, we must equilibrate the datasets to accommodate samples of the attack subtypes. The SMOTE approach oversampled the minority classes, resulting in each attack subtype containing 183,189 samples, thereby achieving a balanced dataset. This results in a total of 1,648,701 samples distributed among the nine classes.



(a) IoTID20 dataset after data augmentation



(b) Edge-IIoTset dataset after data augmentation

FIGURE 7. Balanced class within IoTID20 and Edg-IIoTset datasets: binary classification.

5) FEATURE NORMALIZATION MODULE

The scales of numerical properties frequently demonstrate considerable variance. For example, in the IoTID20 dataset, the range of the "`Flow_IAT_Mean`" feature significantly exceeds that of the "`Fwd_Header_Len`" feature. In this instance, constructing a distance-based classifier with the raw data generates results biased toward the "`Flow_IAT_Mean`" feature. To address this issue, we employed min-max normalization techniques to standardize all attributes within the range of 0 to 1, as represented by Equation 5:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{5}$$

Here, x represents the original value to be normalized, x_{min} and x_{max} denote the minimum and maximum values of the feature, respectively, and x_{norm} represents the normalized value of x. After normalization, the IoTID20 and the Edge-IIoTset datasets are divided into training and testing subsets.

6) DATA SPLITTING

Partitioning data into training and testing subsets is a conventional preprocessing procedure employed to evaluate the efficacy of machine learning and deep learning models [63], [64]. In unbalanced datasets, random partitioning may result in an unequal data distribution, compromising precise model evaluation. To resolve this, we employed a stratified methodology to partition the data into training and testing subsets. A stratified approach divides the dataset into homogeneous subsets [65]. In this study, we used stratified sampling to create a dataset split of 80% for training and 20% for testing. Table 5 shows an overview of the IoTID20 and the Edge-IIoTset datasets that were used. It shows how the data was split between the training, validation, and testing sets.

IV. EXPERIMENTATION AND FINDINGS

We study the simulation results with the IoTID20 and the Edge-IIoTset datasets to gain insights into the intrusion detection performance of the suggested architecture. We assess the effectiveness and efficiency of the proposed method by comparing the obtained performance metrics with established benchmarks and existing intrusion detection techniques.

A. EXPERIMENTAL ENVIRONMENT

We implement and evaluate the suggested approach on the IoTID20 and the Edge-IIoTset datasets using a Raspberry Pi model B, equipped with a quad-core Cortex-A72 (ARM v8) 64-bit SoC operating at 1.5GHz and 8.0 GB of installed RAM. We use Spyder Python (V. 5.5) as our implementation tool, incorporating various libraries consisting of Numpy (V. 1.26.4), Scikit-learn (version 1.3), Pandas (V. 2.2.0), Keras (V. 3.0.4), and TensorFlow (V. 2.15.0). Figure 8 shows our testbed setup of the HIDS-IoMT model and Table 6 summarizes our evaluation environment.

FIGURE 8. The testbed setup of the HIDS-IoMT model.

B. HYPERPARAMETERS SETTINGS IN PROPOSED MODEL

We optimized the hyperparameters of our hybrid CNN and LSTM model to enhance its performance in both binary and multi-class classification tasks for the assessment of our proposed HIDS-IoMT. The binary classification model architecture consists of three Conv1D layers with the ReLU activation function. We employed the Adam optimizer with a learning rate of 0.001 and applied a dropout rate of 0.2 to mitigate overfitting. We conducted training for 50 epochs with a batch size of 128, employing the binary cross-entropy loss function. In the case of multi-class classification, the model architecture had three Conv1D layers and used the Adam optimizer with the same dropout and regularization settings. We chose categorical cross-entropy as the loss function to address the multi-class results. The batch size and the number of epochs constantly stayed at 128 and 50, respectively. As illustrated in Table 7, we carefully adjusted the hyperparameter settings to make the model work better by finding the best balance between efficiency and accuracy. This made it easier to find things in both binary and multi-class classification situations within the HIDS-IoMT framework.

C. EVALUATION METRICS

We evaluated the HIDS-IoMT methodology using accuracy, precision, recall, and the F1-score. This section begins with an analysis of the four key parameters—True Positive (TP), False Negative (FN), False Positive (FP), and True Negative (TN)—used to derive metrics, specifically, accuracy, precision, recall, and the F1-score. True Positive indicates the number of instances correctly classified as normal, while False Negative indicates the rate at which normal data is incorrectly categorized as an attack. False Positive represents malicious instances incorrectly classified as normal, and True Negative counts instances accurately classified as malicious. Each of these evaluation metrics was calculated using Equations 6, 7, 8, and 9.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{6}$$

$$Recall = \frac{TP}{FN + TP} \tag{7}$$

$$Recall = \frac{TP}{FP + TP} \tag{8}$$

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{9}$$

D. BINARY CLASSIFICATION

This section outlines the experimental outcomes of the proposed HIDS-IoMT model, whereby we adjusted hyperparameters across two scenarios: binary and multiclass classification, employing both the IoTID20 and the Edge-IIoTset datasets. The HIDS-IoMT model was examined alongside many established deep learning (DL) and machine learning (ML) models under identical settings, facilitating a comparative examination of its efficacy. Additionally, we demonstrate the model's effectiveness by comparing its performance to relevant studies carried out on these datasets.

Table 8 and Table 9 illustrate that the proposed HIDS-IoMT method outperforms all other approaches in

TABLE 5. Description of IoTID20 and Edge-IIoTset datasets.

Dataset	Feature count	Dataset domain	Publicly available	Label Type	Class count	Training	Testing	Validation
IoTID20	62	IoT	Yes	Binary attack	2	936,548	117,069	117,069
				Multicalss attack	9	1,318,961	164,871	164,871
Edge-IIoTset	46	IoT and IIoT	Yes	Binary attack	2	2,182,396	272,800	272,800
				Multicalss attack	15	16,367,976	2,045,997	2,045,997

TABLE 6. Experimental environment.

Operating system	Linux (Raspbian)
Processor	Broadcom BCM2711, quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz
RAM	8.0 GB
Library	Pandas (2.2.0), Scikit-learn (1.3), Numpy (1.26.4)
Programming Language	Python 3.11.9 64 bit

TABLE 7. Hyperparameters used in our proposed model.

Hyperparameters	Binary classification	Multi-classification
Number of Conv1DLayers	3	3
Number of epoch	50	50
Loss function	binary_crossentropy	categorical_crossentropy
Activation function	ReLU	ReLU
Regularization rate	0.001	0.001
Optimizer	Adam	Adam
Batch size	128	128
Dropout rate	0.2	0.2

TABLE 8. Comparison of binary performance with alternative models and relevant research on IoTID20 dataset.

	IoTID20 dataset				
	Accuracy (%)	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**Test Time (in Sec)**
CNN	99.233	99.381	99.602	99.491	65.3
LSTM	99.384	99.471	99.712	99.591	74.9
[17]	99	97	95	96	233.17
[18]	99.68	99.5	99.53	99.31	11
[19]	98.05	95.35	95.96	95.64	8
[20]	97.25	96.3	95.7	96.5	79.3
Proposed	99.927	99.913	99.99	99.951	81

terms of accuracy, precision, recall, and F1-score. A remarkable 99.92% accuracy is reached by the HIDS-IoMT model, which is better than other machine learning and deep learning methods like CNN, LSTM, and those discussed in [17], [18], [19], and [20]. This result underscores its exceptional capability to accurately distinguish between normal and malicious traffic—a critical feature for IoT applications. Additionally, the HIDS-IoMT model achieves a precision of 99.91%, indicating its effectiveness in minimizing false positives.

TABLE 9. Comparison of binary performance with alternative models and relevant research on Edge-IIoTset dataset.

	Edge-IIoTset dataset				
	Accuracy (%)	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**Test Time (in Sec)**
CNN	99.761	99.898	99.449	99.673	21
LSTM	99.814	99.935	99.549	99.742	44.3
[18]	99.93	99.29	99.29	99.29	27
[19]	99.96	99.83	99.79	99.81	22
[21]	99.994	99.993	99.48	99.4	38
[22]	100	N/A	N/A	N/A	95.04
Proposed	99.941	99.942	99.900	99.921	59.11

Figure 9 and Figure 10 display the training and validation performance, assessing the suggested HIDS-IoMT model for any overfitting issues. From Figure 9a, we found that the proposed model exhibits enhanced performance relative to CNN and LSTM regarding training and validation loss.

The HIDS-IoMT model regularly has the lowest training loss, with values between 0.0007 and 0.0031, signifying highly efficient learning and early convergence. Conversely, the CNN model exhibits elevated training loss, ranging between 0.0078 and 0.0107, indicating reduced stability and a possible challenge in assimilating complex patterns.

LSTM outperforms CNN with training losses ranging from 0.0046 to 0.0060, although it does not attain the same efficiency as HIDS-IoMT. Regarding validation loss, which evaluates the model's generalization to novel data, HIDS-IoMT consistently surpasses both CNN and LSTM, with values between 0.0013 and 0.0040. Despite LSTM demonstrating superior generalization compared to CNN, with validation losses between 0.0056 and 0.0072, it fails to achieve the lower levels reported by HIDS-IoMT.

The identical behavior is evident in the accuracy results, as depicted in Figure 9b and Figure 10b. The results confirm that the proposed HIDS-IoMT model outperforms both CNN and LSTM. The HIDS-IoMT model achieves enhanced training and validation accuracy, indicating its greater ability to learn and generalize from the data. For the most part, the HIDS-IoMT model does better than both CNN and LSTM. Its training accuracy is between 0.9989 and 0.9995, and its validation accuracy is between 0.9969 and 0.9987. Furthermore, CNN exhibits training accuracy values between 0.9891 and 0.9922, whereas validation accuracy spans from 0.9871 to 0.9918, suggesting suboptimal learning and generalization

capabilities. The higher accuracy of HIDS-IoMT shows that it can find threats with more accuracy in real-world situations, making sure that critical systems like the IoMT have reliable intrusion detection.

In addition, experimental results confirm that the increased efficiency of the deployed HIDS-IoMT model has substantial implications for the IoMT. When devices consistently produce large amounts of sensitive health data, the HIDS-IoMT model ensures the accurate and real-time detection of errors and security risks, which is necessary to protect patient safety and data integrity. The model's low training loss shows that it can quickly adapt to the complex and varied data streams that come from medical devices. On the other hand, its low validation loss demonstrates its generality, ensuring reliable performance when applied to new data from various devices in a real-world architecture. This is especially relevant in decentralized architectures featuring fog nodes, where limited computational resources necessitate the use of low-latency, efficient intrusion detection systems.

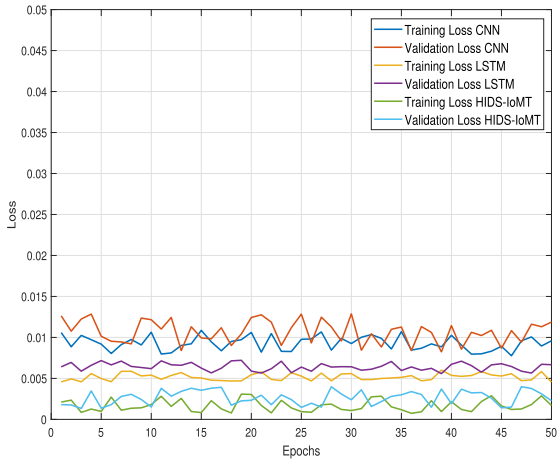
The confusion matrices in Figure 11 and Figure 12 demonstrate that the proposed IDS has shown high detection rates and low false alarms on both the IoTID20 and the Edge-IIoTset datasets.

This demonstrates that our proposed IDS can keep healthcare networks safe from intrusion attempts. This is crucial for maintaining the integrity of sensitive patient data and ensuring the reliability of medical devices. In fact, the identification of 87,987 true positives in Figure 11c, accompanied by a recall rate of 99.99%, underscores its ability to detect practically all potential security breaches.

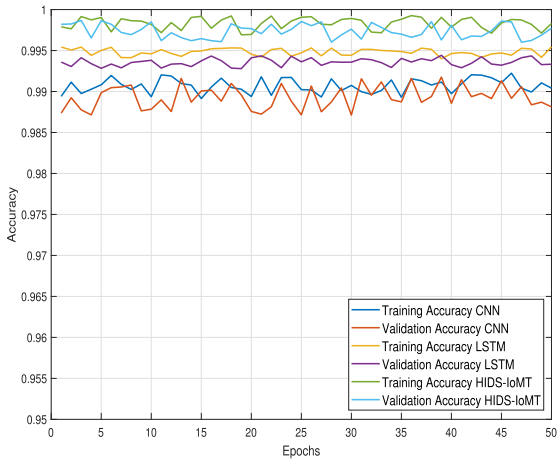
This capability is crucial in healthcare, where even one intrusion could compromise patient safety and data confidentiality. Furthermore, the proposed IDS maintains a low false positive rate, ensuring accurate classification of standard medical activities such as routine data exchanges between devices, thereby minimizing the risk of false alarms that could disrupt essential healthcare operations. Indeed, in healthcare environments, false positives can result in significant consequences, possibly causing system failures or inappropriate administrative actions.

However, despite its exceptional classification accuracy, HIDS-IoMT requires a longer testing duration of 81 seconds compared to other models such as [20] at 79.3 seconds and [19] at 8 seconds. We primarily attribute this to the use of a Raspberry Pi, a device with limited resources, as opposed to more robust platforms used in previous research.

In addition to this, the limited computational power and memory resources of the Raspberry Pi pose significant challenges for executing complex machine learning models. The quad-core ARM Cortex-A72 CPU of the Raspberry Pi operates at 1.5 GHz, significantly slower than desktop-class CPUs or specialized hardware accelerators like GPUs, commonly utilized for training and testing high-performance machine learning models. This leads to extended interpretation durations, especially for deep learning models with numerous parameters, such as HIDS-IoMT, which depends on both CNN and LSTM models for data analysis and classification. Additionally, memory limitations are another significant consideration. The Raspberry Pi, equipped with a maximum of 8.0 GB of RAM, must allocate resources for both the operating system and the model's memory requirements within these constraints.



(a) Loss

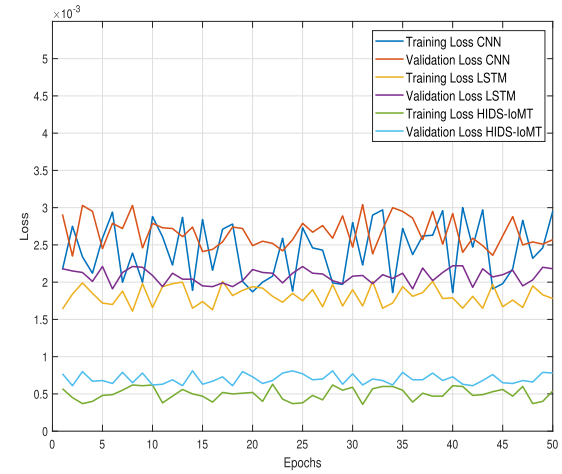


(b) Accuracy

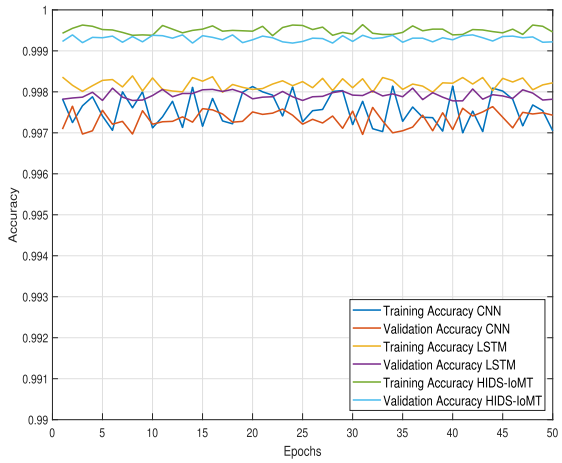
FIGURE 9. Performance of the deployed HIDS-IoMT system during training and validation on IoTID20 dataset.

E. MULTI-CLASS CLASSIFICATION

Along with the binary classification results we already talked about, this section looks at the results of our HIDS-IoMT model's multiclass classification experiments and compares its performance with that of other machine learning and deep learning models. Table 10 and Table 11 indicate that HIDS-IoMT surpasses other leading models in accuracy, precision, recall, and F1-score in multiclass classification results. For example, Table 10 depicts that HIDS-IoMT achieves superior overall performance across all critical criteria, achieving 99.86% accuracy, 99.93% precision, 99.87% recall, and 99.90% F1-score. This confirms the model's capacity for high-accuracy classification and

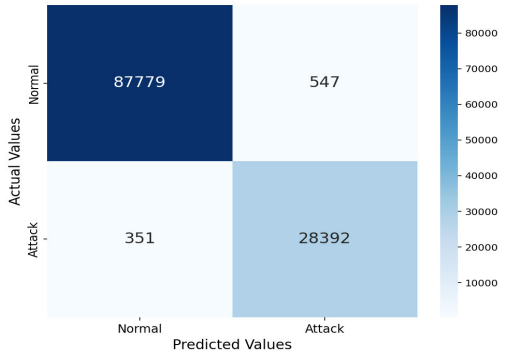


(a) Loss

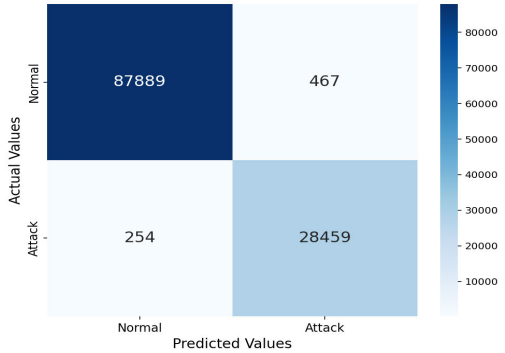


(b) Accuracy

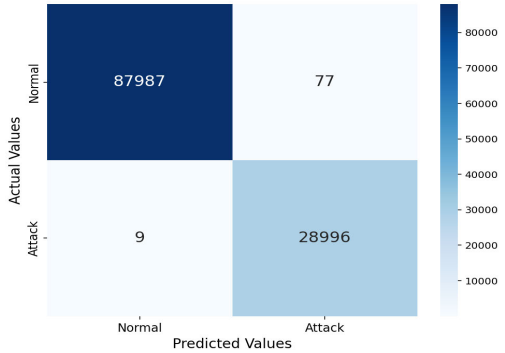
FIGURE 10. Performance of the deployed HIDS-IoMT system during training and validation on Edge-IIoTset dataset.



(a) The convolution neural network (CNN) model



(b) The long short-term memory (LSTM) model



(c) The Hybrid Intrusion Detection System (HIDS) model

FIGURE 11. Confusion matrices for the binary classification of normal and attack scenarios on IoTID20 dataset.

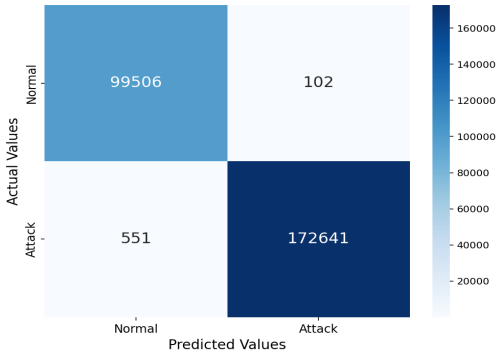
the minimization of false positives, an essential attribute for intrusion detection systems (IDS) in detecting attacks.

Although other models demonstrate stable performance, they do not attain the precision level of HIDS-IoMT. CNN attains an accuracy of 99.09%, [17] gets 91%, while [19] and [20] exhibit inferior performance with accuracies of 96.89% and 98.25%, respectively. Nonetheless, as with binary classification, the unique issue of the HIDS-IoMT model is its extended testing duration of 101.59 seconds, which much surpasses that of models such as [20] (50.9 seconds) and [19] (7 seconds). The longer testing duration is due to the utilization of the Raspberry Pi as a fog node, which comes with constrained computational capabilities.

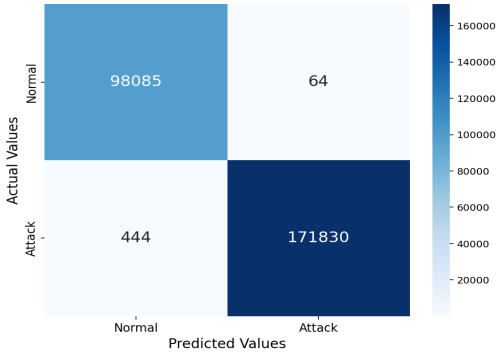
In contrast to the binary classification outcomes, the model demonstrated an extended testing duration for multiclass classification, totaling 101.59 seconds, compared to 81 seconds for binary classification. The increased complexity of multiclass classification explains this minor increase in processing duration. Nonetheless, despite temporal limitations, HIDS-IoMT exhibits strong performance, demonstrating its efficacy in complex IoT systems where incursions may take different forms and require multiclass classification capabilities. Furthermore, implementing our model on a Raspberry Pi as a fog node has numerous benefits. The HIDS-IoMT model consistently produces outstanding outcomes regarding accuracy, precision, recall, and F1-score. In the context of intrusion detection systems, the efficacy in identifying attacks and reducing false positives is crucial.

Despite necessitating more testing time, the excellent classification results obtained on the Raspberry Pi demonstrate its suitability for IoT contexts. These nodes facilitate distributed threat detection with efficient classification while ensuring modest resource utilization, representing an acceptable compromise in such contexts.

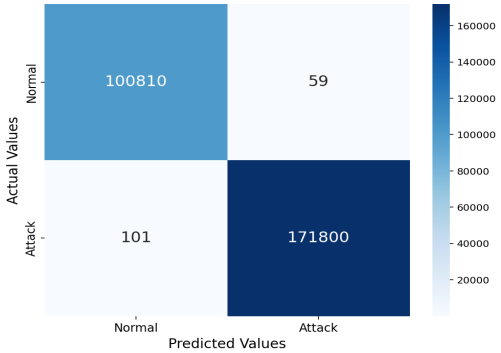
In addition to this, although the accuracy gap between our proposed method and other works across the IoTID20 and



(a) The convolution neural network (CNN) model



(b) The long short-term memory (LSTM) model



(c) The Hybrid Intrusion Detection System (HIDS) model

FIGURE 12. Confusion matrices for the binary classification of normal and attack scenarios in Edge-IIoTset dataset.

Edge-IIoTset datasets may seem minimal, as demonstrated in Table 8, Table 9, Table 10, and Table 11, this finding highlights the difficulties in enhancing highly optimized models, which encompass a variety of complicated attack scenarios. As an example, [18] demonstrates remarkable performance in binary classification with elevated accuracy and F1-scores; nevertheless, it omits multiclass classification, hence constraining its relevance to real-world situations that necessitate the detection of various intrusion forms. In the same way, [19] demonstrates impressive results on the IoTID20 dataset in binary classification, attaining significant accuracy of 98.05%; nevertheless, its efficacy in multiclass classification tasks is comparatively inferior, with an accuracy of 96.89%. Conversely, our suggested hybrid CNN-LSTM architecture consistently attains elevated accuracy, precision, recall, and F1-scores for both binary and multiclass classification tasks, achieving values of 99.860% on the IoTID20 dataset and 99.939% on the Edge-IIoTset dataset for multiclass scenarios. This illustrates the strength and adaptability of our approach across diverse data distributions and attack vectors.

TABLE 10. Multi-classification Performance comparison with other models and related articles in the IoTID20 dataset.

	IoTID20 dataset				
	Accuracy (%)	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**Test Time (in Sec)**
CNN	99.093	99.453	99.349	99.401	94.33
LSTM	99.223	99.595	99.375	99.485	85.5
[17]	91	95	87	85	235.95
[18]	N/A	N/A	N/A	N/A	N/A
[19]	96.89	92.39	87.83	90.05	7
[20]	98.25	97.4	96.8	97.09	50.9
Proposed	99.860	99.935	99.879	99.907	101.59

TABLE 11. Multi-classification Performance comparison with other models and related articles in the Edge-IIoTset dataset.

	Edge-IIoTset dataset				
	Accuracy (%)	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**Test Time (in Sec)**
CNN	99.760	99.834	99.799	99.816	90
LSTM	99.764	99.834	99.801	99.818	82
[18]	N/A	N/A	N/A	N/A	N/A
[19]	99.95	99.46	99.61	99.53	23
[21]	N/A	N/A	N/A	N/A	N/A
[22]	99.64	N/A	N/A	N/A	219.29
Proposed	99.939	99.948	99.952	99.950	92.5

Figure 13 and Figure 14 present the confusion matrix for the hybrid algorithm applied to multiclass classification in the IoTID20 and the EdgeIIoTset datasets. The proposed intrusion detection system exhibits considerable efficacy in protecting IoT healthcare systems from Denial-of-Service (DoS) attacks, such as SYN flood attacks.

Eventually, higher diagonal values in the confusion matrices indicate that the model has executed a greater number of correct predictions. Moreover, the confusion matrices for both datasets distinctly demonstrate that the proposed system has correctly identified different types of intrusion threats, proving that it works for trust-based Internet of Medical Things intrusion detection. These outcomes reinforce our observations that our proposed IDS ensures protection for healthcare systems against DoS attacks while preserving the continuous operation of medical devices and systems. This underscores its suitability for environments where security, reliability, and uninterrupted service are critical.

V. DISCUSSION

Experiments confirm that the deployed HIDS-IoMT model is very efficient in detecting malicious activity within IoMT networks. It achieves superior accuracy, precision, recall, and F1-score compared to other approaches for both binary and

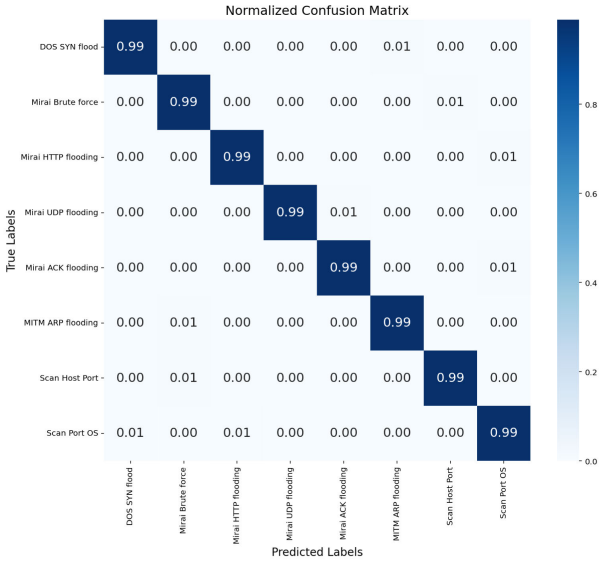


FIGURE 13. Confusion matrix of the proposed HIDS-IoMT scheme on IoTID20 dataset.

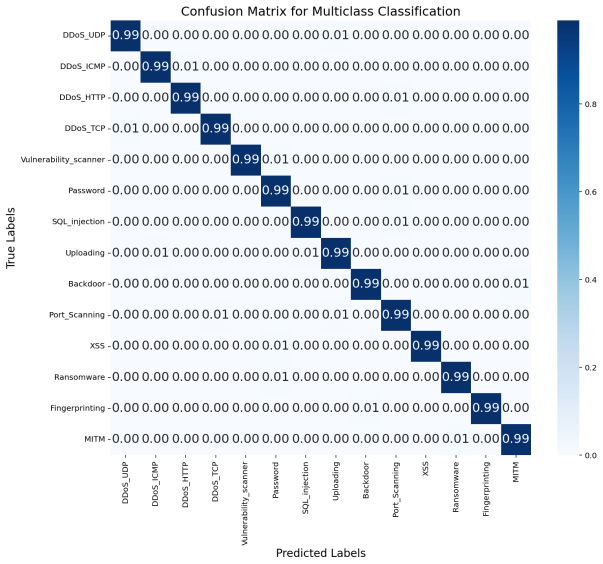


FIGURE 14. Confusion matrix of the proposed HIDS-IoMT scheme on EdgeIIoTset dataset.

multiclass classification tasks. Indeed, the modular design of our proposed framework facilitates adaptation to extensive IoT ecosystems, encompassing Industrial IoT (IIoT), smart cities, and connected home applications, as these settings display dynamic traffic patterns and are susceptible to various cyber threats. The main structure of our proposed framework, which includes Convolutional Neural Networks (CNNs) for feature extraction and Long Short-Term Memory (LSTM) networks for modeling temporal sequences, can also be easily changed to work with different types of networks. In addition to this, the selection of the IoTID20 and Edge-IIoTset datasets utilized in this study is twofold: First, the IoTID20 dataset contains real-world IoT traffic data from a variety of attack scenarios, such as DoS, Mirai Botnet, and ARP spoofing. This provides a solid base for testing the model's ability to identify a wide range of attacks. Second, the Edge-IIoTset dataset includes IIoT traffic and a wide range of 14 attack types, including both common and advanced attacks. This makes it a great representation of IoT and IIoT environments in the real world.

The fog computing architecture presented in this paper is crucial and addresses the latency-sensitive requirements of healthcare systems. In medical IoT environments, delays in detecting and responding to intrusions can compromise patient safety and hinder critical healthcare services. Also, our suggested system processes data close to IoMT devices using fog computing architecture. This greatly reduces latency and ensures quick responses to any threats that might be present. Additionally, the use of inexpensive Internet of Things (IoT) devices like Raspberry Pi as fog nodes shows how useful our method is for enabling resource-efficient intrusion detection that is not dependent on centralized cloud systems. Also, the modular structure of our algorithm makes it easier to update and scale, protecting healthcare applications that are sensitive to latency and limited in resources in a strong and cost-effective way.

The experiments and the observations reveal that the proposed framework was influenced by the constraints of limited hardware resources, resulting in testing durations of 81 seconds for binary classification and 101.59 seconds for multiclass classification. Nonetheless, the model's precision of 99.91% in binary classification effectively minimizes false positives, ensuring minimal disruption to healthcare operations where false alarms can have significant consequences. The experimental results also confirm a low validation loss and enhanced generalization capacity, emphasizing the model's remarkable adaptability to new data—a critical attribute given the dynamic nature of cyberthreats in IoMT ecosystems.

VI. CONCLUSION AND FUTURE WORKS

This research presents a hybrid deep learning-based intrusion detection system (HIDS-IoMT) designed to enhance the security of Internet of Medical Things (IoMT) networks against advanced cyber threats, specifically Distributed Denial of Service (DDoS) attacks. The HIDS-IoMT model combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to accurately find valuable behavior. Implementing this model in a fog computing architecture on a Raspberry Pi facilitates decentralized processing near IoMT devices, thereby minimizing latency and improving responsiveness. We conducted an experimental evaluation on the IoTID20 and the Edge-IIoTset datasets, demonstrating the model's superior performance with over 99% accuracy, precision, recall, and F1-score in both binary and multiclass classification tasks. This demonstrates the model's ability to effectively distinguish between normal and anomalous traffic, a critical characteristic for the dynamic and high-stakes contexts in medical IoT. Despite its prolonged processing time resulting from hardware constraints on the Raspberry Pi, the HIDS-IoMT model shows promise

for effective implementation in practical scenarios where precision and reliability are critical. Future research will concentrate on enhancing computational efficiency to minimize latency, particularly in multiclass situations, and on extending the model to encompass wider IoT applications. The goal is to make the proposed model more useful in a wider range of IoT ecosystems by looking into ways to improve threat intelligence and make it easier to adapt to new attack vectors, especially in places with limited resources.

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